January 16, 1963

SUPERSEDING
Fed. Spec. W-C-256a (In Part)
November 26, 1952

FEDERAL SPECIFICATION

BATTERY CHARGING EQUIPMENT; MOTOR-GENERATOR, CONVERSION AND NON-CONVERSION TYPE, (FOR CHARGING INDUSTRIAL STORAGE BATTERIES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers charging equipment (motor-generator type) for charging industrial storage batteries used in materials-handling equipment.

1.2 Classification.

1.2.1 *Types and sizes.* This specification covers single-circuit, self-contained, motor-generator battery charging equipment. Unless otherwise specified, motor-generators shall consist of one of the following types and sizes (6.2):

Type I.—Chargers for nominal 36-volt batteries in the following sizes, categories, capacities, and discharge rates:

Size A.—Lead-acid, 18-cell category with a range of capacities from 300 to 750 ampere-hours at the 6-hour discharge rate.

Size B.—Nickel-iron, 30-cell category with a range of capacities from 280 to 750 ampere-hours at the 5-hour discharge rate.

Size C.—Combination for sizes A and B.

Type II.—Chargers for nominal 12-volt batteries in the following sizes, categories, capacities, and discharge rates:

Size A.—Lead-acid, 6-cell category with a range of capacities from 250 to 600 ampere-hours at the 6-hour discharge rate.

Size B.—Nickel-iron, 10-cell category with a range of capacities from 250 to 600 ampere-hours at the 5-hour discharge rate.

Size C.—Combination for sizes A and B.

Type III.—Chargers for nominal 12-volt batteries in the following sizes, categories, capacities and discharge rates:

Size A.—Lead-acid, 6-cell category with a range of capacities from 600 to 900 ampere-hours at the 6-hour discharge rate.

Size B.—Nickel-iron, 10-cell category with a range of capacities from 600 to 900 ampere-hours at the 5-hour discharge rate.

Size C. Combination for sizes A and B.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the

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issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

CC-M-641—Motors, Alternating-Current (Integral Horsepower).

FF-B-171—Bearings, Ball, Annular (General Purpose).

QQ-S-698—Steel. Sheet and Strip, Low-Carbon.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-16298—Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Electrical Manufacturers' Association Publication:

MG-1-1959—Motors and Generators.

(Application for copies should be addressed to National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, N. Y.).

Underwriters' Laboratories, Inc. Standard:

UL508—Standard for Industrial Control Equipment.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York, 207 East Ohio Street, Chicago, Ill., or 1655 Scott Blvd., Santa Clara, California.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each bidder shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. Standard UL508. The label or re-examination listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

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3.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that the equipment conforms to the requirements, including methods of tests, of UL508.

3.1.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.2 **General.** Motor-generator battery charging equipments covered by this specification shall be complete with all necessary devices and controls for automatic operation when delivered, requiring only connections to the associated electrical supply. They shall be constructed so that they may be installed side by side where compact installation of a multiple charging station is desired. Structurally, the motor-generators shall be designed so that they may be readily moved from one location to another through the use of suitable accessories and shall be capable of continuous operation at maximum rated loads. All major and minor assemblies, including accessories, shall be new and of the manufacturer's standard design or model current at the time of delivery, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturer shall be so established in the industry that prompt and continuing service and delivery of spare parts may be assured.

3.2.1 *Preproduction model.* When specified (6.2), a preproduction model shall be submitted for examination, test, and inspection in accordance with section 4, and such other tests as are necessary to determine compliance with the provisions of this specification and the contract. In all respects not specifically covered by this specification, the preproduction model shall represent the identical quality of materials, workmanship, and design which will be used in the fabrication of the production units. Acceptability shall be based on an evaluation of the design, structural, and operational characteristics relative to safety, operational efficiency, and standard design practice. Any fabrication of contractual requirements without approval of the preproduction model shall be at the contractor's risk. The approved preproduction model shall be used as a guide to inspect production units and may be presented for acceptance as the last unit of the contract. Examination and testing, whenever practicable, will be conducted at the contractor's plant.

3.2.2 *Material.* Material shall be as specified herein. Material not definitely specified may be manufacturer's normal commercial product or the standard commercial component(s) used in the fabrication of the manufacturer's normal commercial product. "Normal commercial product" shall be interpreted to mean an end item produced by a manufacturer engaged in the commercial production of end item(s) covered by this specification. The Government reserves the right to require any bidder to submit, for approval of the contracting officer, detailed engineering drawings of proposed end items or components differing from the "normal commercial products or standard component(s)" prior to and as a condition precedent to the award of the contract.

3.2.2.1 *Insulation.* Insulating materials shall be mechanically strong and able to resist any pressure, tensile strains, bending, bruising, or shock vibration likely to be encountered in service. Insulating materials shall be capable of withstanding the specified temperatures and conditions without deterioration.

3.2.2.2 *Electrical-resistance alloys.* Electrical-resistance alloys shall be heat-resistant and corrosion-resistant.

3.2.2.3 *Structural steels.* Structural steels

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shall conform to the applicable and pertinent requirements of QQ-S-698.

3.2.3 *Design.* Charging equipment covered by this specification and components incorporated therein shall conform to accepted design practices and shall insure safe, efficient, and economical operation under normal service conditions. There shall be no evidence of excessive failures when operated under normal conditions.

3.3 Construction.

3.3.1 *General.* Motor-generators shall be self-contained, consisting of an alternating-current motor-direct-current generator set; motor control and direct-current charging control. They shall be of the two bearing unit, vertical or horizontal type, operating at speeds of not more than 3600 revolutions per minute maximum and shall be designed having a drooping voltage load characteristic to produce an equivalent reduction in the charging current as the battery voltage increases with the state of charge. Motor-generators shall conform to the applicable requirements of the National Electrical Manufacturers' Association standards for motors and generators (MG-1-1959) and all established recommended standards as adopted by the Industrial Truck Association.

3.3.1.1 *Balance.* The rotating assembly (motor rotor and generator armature) shall be statically and dynamically balanced at all speeds up to 110 percent of rated normal synchronous speed.

3.3.1.2 *Overspeed limitations.* Motor-generator sets shall be capable of withstanding stresses and strains induced when subjected to 25 percent overspeed.

3.3.1.3 *Noise.* Machines shall be free from excessive magnetic noise, lamination or teeth buzzing noises, screech of brushes, and pneumatic howling of ventilating system. This is not intended to describe a unit especially designed for "noiseless" operation.

3.3.1.4 *Supply voltage.* Unless otherwise specified in the invitation for bids (6.2), charging equipment covered by this specification shall be capable of operating under the following nominal voltage supply characteristics:

Type I—208 volts, 3 phase, 60 cycle.

Type II—208 volts, single phase, 60 cycle.

Type III—208 volts, 3 phase, 60 cycle.

Charging equipment shall be capable of operating satisfactorily at plus or minus 10 percent of rated line voltage.

3.3.2 *Motor.* Unless otherwise specified in the invitation for bids (6.2), the driving motor shall be squirrel-cage induction, alternating-current type with normal starting torque and low starting current conforming to the applicable requirements of CC-M-641. It shall have sufficient close speed regulation to permit the generator characteristics as specified herein and with power supply characteristics as defined in 3.3.1.4. The alternating-current motor shall be capable of driving continuously the generator to which it is connected at its full rated capacity with temperature rise not to exceed NEMA standards for open type construction machines at rated primary voltage. Motors other than open type construction shall conform to the applicable NEMA standards as to temperature rise.

3.3.2.1 *Rating.* The horsepower rating of the motor shall be such that it shall continuously drive the generator at rated full load without exceeding the allowable temperature rise as specified in NEMA standards.

3.3.3 *Generator.* Generators shall be of the continuous-duty type, direct-current, shunt or compound wound. The voltage characteristics of the generator shall be such as to provide suitable charging rates for the time specified. All generators shall conform to the applicable NEMA standards.

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3.3.3.1 *Rating.* Output rating of the generator shall be rated in kilo-watts, or amperes and direct-current volts.

3.3.3.2 *Commutator.* The generator commutator shall be designed having adequate strength to withstand any load up to 25 percent overload without destructive sparking. Commutator bars shall be made of hard-drawn copper with the segments accurately spaced and aligned, separated by mica insulation of uniform thickness. Commutator mica shall be properly undercut forming grooves of not less than 3/64-inch minimum.

3.3.3.3 *Brushes.* Brushes shall be carbon or a combination of carbon and graphite or metal. They shall be designed with sufficient capacity to prevent undue heating when the generator is carrying rated load. Brushes shall be carefully ground to fit commutator prior to shipment. Each brush shall be provided with a heavy flexible shunt or pigtail, properly secured. Brushes shall be of the greatest tangential width consistent with good commutation, and of relatively small height to assist in creating stability and freedom from chatter.

3.3.3.3.1 *Brush holders.* Brush holders shall be designed with accuracy and rigidly mounted to prevent brush binding. The brush rigging shall be of rigid construction so as to reduce vibration at the brushes to a minimum. Means shall be provided to permit adjustment of the brush position and prevent loosening or shifting. Holders shall be easily accessible for adjustment and renewal of brushes and springs. Provisions shall be made for adjusting the brush spring tension without the use of tools. Springs shall not carry current. The correct brush setting shall be plainly indicated by permanent marks on the brush holder yoke and the generator frame or end bracket.

3.3.3.4 *Direction of rotation.* The direction of rotation of the generator shall be clearly indicated by a nameplate.

3.4 **Auxiliary equipment.** The auxiliary equipment of each charger shall include the following: Alternating-current magnetic contractor or suitable means for disconnecting the alternating-current supply circuit; an electrically operated interval timer switch; voltage responsive device in addition to timer switch for automatic charge termination of lead and nickel-iron-alkaline batteries; battery circuit contactor; suitable means for selective charging of lead and nickel-iron batteries, when required; fuse or circuit-breaker protection in one line of the battery circuit; overload or temperature protection which will open the alternating-current line circuit.

3.4.1 *Charging panels.* All motor-generator control and battery charging circuit panels for chargers shall be of the metal enclosed dead-front type construction with instruments and operating controls mounted externally to permit normal operation without physical exposure to live parts and shall be easily accessible for inspection. The sides, back, and top may be enclosed with expanded metals. Motor-generator controls may be mounted in dead-front enclosure on the unit. Panels shall include the motor contactor, battery contactor, voltage responsive device, field rheostat or resistor, ammeter, overload reset button, timer, necessary control relays, contactors, and fuse clips. The control panel(s) housing all control components, shall be physically located for maximum convenience to the operator. All controls shall be easily operated, even if the charger should be placed on a stand.

3.4.1.1 *Panel frames and enclosures.* Panel frames shall be made of angle iron, tee iron, or other rolled or formed steel shapes. Sheet steel for enclosures shall be not less than No. 18 U.S. Standard gauge (0.0490 inch).

3.4.2 *Generator field rheostat.* The generator field rheostat, when used, shall be so designed that the generator voltage may be readily adjusted to the normal specified

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value, within plus or minus 0.5 percent. For normal operation, the field rheostat shall be fixed in one position. The field rheostat shall be equipped with provision for being clamped permanently in one fixed position subject only to adjustment by a competent operator. Rheostats are not required if adjustable resistors are provided.

3.4.3 *Ballast resistors.* Ballast resistors, when used, shall be of the cast-grid type, flat ribbon-wound type or edgewise wound type. They may be readily adjustable by suitably designed connections which shall be secured at any desired resistance value within the limit required. They should be so mounted as to be easily accessible and to shield other panel parts from excessive heat.

3.4.4 *Motor starter.* The motor starter shall be of the across-the-line, full voltage, magnetic type, conforming to UL508 for Industrial Control Equipment, and shall provide thermal overload and low voltage protection.

3.4.5 *Meters.* Chargers shall be equipped with a direct-current ammeter having a minimum range of 0-200 amperes and a direct-current voltmeter, when required, having a minimum range of 0-75 volts. Meters shall be a high-grade d'Arsonval type, of good quality with an initial accuracy of plus or minus 2 percent of full-scale reading and shall have a radial-dial scale not less than 3 inches in diameter, plus or minus 1/4 inch.

3.4.6 *Charge controls.* The voltage relay time-switch controls shall be so constructed that they may be readily set to the predetermined time. All time-switch controls shall be interlocked with the motor starter so a power-supply failure will stop the timer and, upon restoration of the supply, start it again.

3.4.6.1 *Voltage responsive device.* Chargers shall be equipped with a voltage responsive system for automatic charge termination on both lead-acid and nickel-iron batteries.

3.4.6.1.1 *Lead-acid batteries.* The voltage responsive system for charging lead-acid batteries shall be adjusted to operate when the voltage rises to 2.37 volts per cell within a tolerance of plus or minus 1 percent as measured at the battery terminals at an ambient temperature of 25°C. The voltage responsive device shall be inversely temperature compensated to cause a change in operating voltage of between 0.003 and 0.004 volts per cell per degree Fahrenheit between the limits of 0°C. to 40°C. An electrically operated, manually preset timer switch responsive to the operation of the voltage relay shall be included. The operation of the voltage responsive device shall serve to start the timer switch and after an adjusted interval of 2.5 or more hours, the opening of switch contacts shall de-energize the charging circuit contactor, thus terminating the charge. The voltage responsive system shall include the complete circuit from the battery to the voltage responsive device, including all of the following (when used) : Switches, multiplier resistors, lead or cables, connectors, protective devices, etc.

3.4.6.1.2 *Nickel-iron batteries.* The voltage responsive system for charging nickel-iron batteries shall be adjusted to operate when the battery voltage rises to 1.695 volts per cell within a tolerance of plus or minus 0.005 volts as measured at the battery terminals at an ambient temperature of 25°C. The voltage responsive device shall be inversely temperature compensated so as to cause a change in operating voltage of between 0.002 and 0.003 volts per cell per °F. between the limits of 0°C. to 40°C. A manually preset, electrically operated timer switch responsive to the operation of the voltage relay shall be included. The operation of the voltage responsive device shall serve to start the time switch, and when the adjusted time switch has reached the end of its preset timing interval of one hour, the opening of its contacts shall function to terminate the

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charge. The relay shall be sturdy, sensitive, and have repetitive accuracy. Suitable cushioning shall be provided so that the operating characteristics will not be affected by mechanical shock or vibration.

3.4.6.1.3 *Interval timer.* The timer shall be of the manually preset type with a range of at least 0-3 hours. The operation of actuating the timer for the number of hours desired will automatically start a built-in motor of the synchronous or shaded pole type, which through a train of gears will drive the indicating knob or lever of the timer toward zero. When the indicating knob or lever returns to zero, a built-in switch will be actuated which in turn will either operate relays or contactors to or directly terminate the battery charge. Motor and gears of the timer shall be covered so as to be dusttight, and shall be so designed that routine service and lubrication is unnecessary during the life of the equipment.

3.4.7 *Battery leads.* Each charger shall be supplied with a 10-foot flexible lead or leads of a size to conform to UL508 for Industrial Control Equipment, and shall be complete with lugs on one end and charging connector(s) on the other end.

3.4.7.1 Unless otherwise specified in the invitation for bids (6.2), the charging connector(s) shall be the charging half of the standard type EC connector of the Industrial Truck Association, of a size sufficient to accommodate the battery leads.

3.4.8. *Terminal lugs.* All control wiring shall be provided with solder or crimped terminal lugs.

3.4.9 *Lifting means.* Provisions shall be made for the hoisting of motor-generator units by means of lifting eyes, lugs, or equivalent.

3.5 Performance.

3.5.1 *Selective charging.* A simple toggle switch or other suitable means (when required) shall be provided to readily convert from nickel-iron charging to lead-acid battery charging or visa-versa for conversion-type charging equipment.

3.5.2 *Charging method.* The method of charging employed in charging both lead-acid and nickel-iron batteries shall be left to the discretion of the manufacturer of the equipment, as long as the method used and the design of the equipment meets the requirements outlined in 3.5.2.1 and 3.5.2.2.

3.5.2.1 *Lead-acid batteries.* Charging equipment shall be capable of completely charging a fully discharged lead-acid type battery in the range of sizes specified, within 8 hours or less. A fully discharged lead-acid battery is defined as one in which the volts per cell is 1.75 volts, at a discharge load in amperes of 1/6 the ampere-hour rating of the battery with electrolyte temperature of 25°C. The starting rate of charge shall normally be 4 to 6 times the finish rate. The finish rate shall not exceed 5 amperes per hundred ampere-hours of rated capacity of the battery to be charged.

3.5.2.2 *Nickel-iron-alkaline battery.* Charging equipment shall be capable of completely charging a fully discharged nickel-iron type battery in the range of sizes specified within 7 hours or less. A fully discharged nickel-iron battery is herein defined as one in which the volts per cell is 1.00 volt, at a load in amperes of 1/5 the ampere-hour rating of the battery with the electrolyte temperature of 25°C. The initial starting rate shall not exceed 125 percent and the final rate shall be not less than 80 percent with the average rate for the entire charge equal to the normal rate of the battery being charged. Normal rate is defined as 1/5 the rated ampere-hour capacity of the battery at the 5-hour rate.

3.5.3 *Operation.* Motor-generator sets shall have sufficient capacity and proper flexibility of control to properly charge industrial storage batteries as outlined in 1.2.

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Upon connecting a battery charging circuit, the operation of the charge-start device shall automatically start the generator set. The battery shall continue on charge until terminated prior thereto by operating the individual "stop" device. At the end of the preset time, or when the clock reaches zero hours, it shall de-energize the magnetic coil of the across-the-line motor starting switch, shutting down the set, disconnecting the alternating-current power from the time clock, and opening the battery charging circuit. In case of failure of power supply, the motor starter shall open, and its electrical interlock shall function to cause the opening of the charging circuit contactor, thus providing reverse-current protection. Upon resumption of the power supply, the starter shall reconnect the motor and when the generator voltage again builds up to the proper value, the equipment shall reconnect the batteries, resuming the charge.

3.5.3.1 *Temperatures.* The temperature rise, above the temperature of the cooling medium for each of the various parts of the generator set shall not exceed the values recommended by NEMA for the type of insulation used.

3.5.3.2 *Dielectric strength.* The insulation of the motor-generator set shall be capable of withstanding without breakdown the dielectric test outline in 4.6.5.

3.5.3.3 *Efficiency.* Efficiency of the motor-generator set shall be based on the ratio of the DC power output to the AC power input measured at the charger terminal in such a manner as to exclude DC cable losses and shall be not less than 70 percent when delivering rated output.

3.6 **Bearings.** Motor and generator bearings shall be of the oil lubricated or lifetime sealed grease lubricated ball or taper roller type in accordance with the applicable requirements of FF-B-171. When oil-lubricated sleeve-type bearings are provided, convenient means for determining oil level and examining oil rings in operation shall also be provided.

3.7 **Lubrication.** Readily accessible means shall be provided for applying and, where necessary, checking the level of lubricant in all parts and components.

3.8 **Finish.**

3.8.1 *Treatment and painting.* All exterior surfaces of the units shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust; and shall be painted as soon as practicable after cleaning. Not less than one coat of metallic-base primer and one or two coats of synthetic enamel shall be applied to insure protection against rust and corrosion when operated in damp atmosphere or under adverse transportation or storage conditions. Primer and finish coats shall be applied in accordance with best commercial practices to assure complete coverage and durability of the finish. Unless otherwise specified in the invitation for bids (6.2), the finish color shall be in accordance with manufacturer's normal commercial practice. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, or other defects. Surfaces of components and assemblies not normally painted with a finish coat as specified herein, shall be cleaned and prepared in accordance with good commercial practice.

3.9 **Marking for identification.**

3.9.1 *Polarity.* The polarity of all direct-current wiring shall be plainly and durably indicated.

3.9.2 *Nameplate.* Each unit shall be identified by stamped, engraved or etched nameplate (s) of corrosion-resistant metal in accordance with standard commercial practice. Nameplate(s) shall be in a visible and safe location.

3.9.2.1 *Alternating-current motor.* Motor nameplate shall be marked with man-

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ufacturer's type, horsepower, output, time rating, normal temperature rise, revolutions per minute, frequency, number of phases, normal voltage, and full load amperes.

3.9.2.2 *Direct-current generator.* Generator nameplate shall be marked with the manufacturer's type designation and frame number, the kilowatt or ampere output capacity, voltage rating and temperature rise.

3.9.2.3 *Panels.* The manufacturer shall furnish and affix a corrosion-resistant metal nameplate in an easily visible and safe location on the control panel in accordance with standard commercial practice.

3.9.3 *Components.* Components such as magnetic contactors, circuit breakers, relays, charge controls, motor starters, etc., shall be completely identified to facilitate replacement.

3.10 **Protective features.**

3.10.1 *Fuses.* The direct-current output circuit of the charger shall be provided with overload protection or readily accessible and easily replaced fuses of the renewable cartridge type. The continuous current-carrying capacity of fuses shall not exceed 125 percent of the maximum starting current.

3.10.2 *Thermal control.* Thermal control or other suitable approved devices shall be incorporated to protect chargers against burnout from over-heating. Thermal overloads in the motor starting contactor will be permissible.

3.10.3 *Automatic controls.* Each charger shall include the following automatic features: Shutting down of the charging equipment when the charge is completed and opening of the battery charging circuit on completion of charge or in case of line interruption; restarting the charging equipment and closing of the battery circuit upon resumption of power supply following an interruption; control of the charging rate for any system employed in the design of the charging equipment.

3.11 **Interchangeability.** Corresponding units and replaceable assemblies, sub-assemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are replaceable shall be as determined by the contracting officer.

3.12 **Workmanship.** Finished units shall be clean, well made, and shall be free from any defects which may affect appearance or serviceability. All edges shall be smooth, rounded, and free from burrs or other irregularities. Unnecessary pores, pockets, reservoirs, etc., where oil, water and dirt might collect, shall be eliminated. Welded joints shall be round, smooth, free from pits, holes, fissures, and rough projecting edges and shall be thoroughly fused to the base metal.

3.12.1 *Castings.* Castings shall be of best quality normally used in commercial practice and shall be of uniform quality and conditions, free from injurious blowholes, porosity, hard spots, shrinkage defects, cracks, misplaced coring, or other injurious defects. Castings shall be commercially machinable and shall not be repaired, plugged or welded to adversely affect the strength, use or machinability.

3.13 **Maintenance, service, repair, and parts manuals.**

3.13.1 *Technical manuals.* One copy of a technical manual shall be furnished by the contractor with each charger at time of shipment. The manual shall be single-volume or supplementary and shall have content as noted below.

(*a*) General description. The manual

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shall be applicable to each make and model of the equipment furnished under the contract, including wiring diagrams, etc., and adequate instructions for the maintenance and repair of the equipment. The manual shall have the appropriate contract number clearly indicated on the cover, or an inside page, or by fly-leaf insertion:

1. Range of "Serial Numbers" covered.

2. Any other identification data necessary to relate this manual to the applicable equipment.

(*b*) Parts list or parts breakdown. The Parts Section shall cover the complete equipment being furnished, including all accessories, auxiliary components, and attachments; and shall include a component breakdown of all assemblies and subassemblies. Each accessory auxiliary attachment, assembly, subassembly and part shall be identified by a "Part Number", except common hardware-type items such as nuts, bolts, washers, keys and similar items which may be identified by description. Each "Part Number" shall provide positive identification for procurement, stocking, and issue purposes without reference to equipment serial numbers, equipment model numbers, specific parts lists, drawing or symbol numbers, nomenclature special coding, or any other embellishment. Manufacturer's commercial manual will be acceptable providing they meet the above requirements. If they contain additional information which is not pertinent to the specific equipment being purchased, such information shall be deleted or clearly defined. Manuals shall be attached to the equipment. Additional distribution of above manuals shall be as specified in the invitation for bids (6.2).

3.14 Military requirements.

3.14.1 *Finish.* For Navy Department purchases, the finish color shall be color No. 16187 of Fed. Std. No. 595. For Department of Army purchases, the finish color shall be as specified in the invitation for bids (6.2).

3.14.2 *Nameplate.* For military purchases, the nameplate shall be in accordance with MIL-STD-130.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection.

4.2.1 *Preproduction model.* When a preproduction model is required, inspection and testing shall be made of a completed unit in accordance with test procedures outlined herein for conformance to the requirements of this specification. Test and inspection of the preproduction model shall be under the supervision of the cognizant Government representative in accordance with inspection procedures contained in table I. Written approval of the preproduction model by the authorized Government representative shall not relieve the contractor of his obligation to supply equipment conforming to this specification or other requirements set forth in the contract. It shall be understood that such approval is, in all cases, secondary to satisfactory operation under normal service conditions.

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TABLE I.

Inspection	Rqmt. para.	Test para.	Preproduction	Group A AQL	Group B sampling plans
Workmanship, visual and mechanical, major	3.12	4.4	X	1.5 DPHU*	—
Workmanship, visual and mechanical, minor	3.12	4.4	X	6.5 DPHU*	—
Interchangeability	3.11	4.5	X	—	10% AQL L-6 N & T L-4 reduced
Operation of controls	3.4.6	4.6.1	X	1.0%	—
No-load power input	3.3.1.4	4.6.2	X	—	—
Dielectric test	3.5.3.2	4.6.5	X	1.0%	—
Regulation	3.5.2.1 3.5.2.2	4.6.3	X	—	6.5% AQL L-7 N & T L-5 reduced
Heating test	3.5.3.1	4.6.4	X	—	—

* Defects per hundred units.

4.3 Inspection for acceptance. Inspection in general shall consist of group A and group B inspection as specified in 4.3.1 and 4.3.2. Group B inspection shall be performed on lots or samples that have passed group A inspection. Determination of acceptability of product as to compliance with the requirements of this specification shall be in accordance with procedures set forth in MIL-STD-105, except as otherwise specified.

4.3.1 *Group A inspection.* Group A inspection shall be as specified in table I and in accordance with MIL-STD-105. The acceptance quality levels (AQL's) shall be level II. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

4.3.2 *Group B inspection.* Group B inspection and sampling plans shall be as specified in table I and in accordance with MIL-STD-105. Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.4 Visual and mechanical inspection. Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finish, etc., shall be inspected for workmanship. Markings and workmanship shall be examined for compliance with this specification. Examination also may be made for other visual and mechanical defects similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105.

4.5 Inspection for interchangeability. The equipment shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.11. The following should be considered when applicable: (1) External covers and insertable assem-

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blies when such affect mating parts; (2) location of separate parts or assemblies which must mate, such as cases, covers, and mountings; and (3) location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts or plug-in assemblies and major units; and location of mating parts on the plug-in assembly or major unit.

4.6 Tests. In addition to the test specified herein, the Government reserves the right to use such additional information or methods deemed necessary to determine compliance with the requirements of this specification.

4.6.1 *Operation of controls.* Automatic control panels shall be tested to insure that they function as intended. Operational tests shall be made to establish the fact that all voltage relays are properly adjusted, time switches operate as intended, and that ballast resistors, when used, have been calibrated for the duty intended. Meters may be tested for tolerances indicated herein prior to installation on the control panel.

4.6.2 *No load power input.* With no load connected, the power input to the motor-generator shall be measured at rated input voltage.

4.6.3 *Direct-current generator voltage regulation.*

4.6.3.1 *Shunt-wound generators.* The regulation of machines designed to give an inherent drooping voltage characteristic shall be such that the machine, when heated to correspond with the conditions at the end of charge, in an ambient temperature of between 20°C. and 30°C. will develop a voltage of at least 2.5N (where N equals number of lead-acid cells, series connected), at a current equal to the correct finish rate of the battery for which the machine is designed. Machines designed for use with nickel-iron batteries, shall be given a "cold" start-charge adjustment enabling the generator to develop a current output equal to 120 percent of normal 5-hour discharge rate at a minimum voltage of 1.6 volts per cell. The regulation shall then be such that the machine, when heated to correspond with the conditions at end of charge in an ambient temperature of between 20°C. and 30°C. will develop a maximum voltage of 1.8 N at a current equal to 80 percent of the battery finish rate, for which the machine is designed.

4.6.3.2 *Compound-wound generators.* Test for voltage regulation shall be made in an ambient temperature between 20°C. and 30°C. with the starting temperature of the machines within this range. Start at 125 percent of the rated generator capacity. Reduce the load in five approximately equal steps to five percent of full-rated capacity within 15 minutes. Then increase the load to 100 percent of full-rated capacity in approximately four equal steps within 15 minutes. The terminal voltage shall be read at each loading. The voltage regulation will be the maximum difference between the voltage at full load and five percent load at the start and end of test divided by the maximum voltage and expressed in percent. This voltage regulation shall be not more than plus or minus 3.0 percent.

4.6.4 *Heat-run test.* When the generator has been properly prepared, adjusted for proper commutation with well fitted brushes, and when the thermometers have been properly placed on bearing housings, stationary windings, pole shoes, stator iron, and exhaust air paths, the normal load heat run may be started (compound-wound machines shall be adjusted for the required voltage regulation). Normal load is the actual load as encountered during a normal battery charging cycle for the largest battery to be charged. Normal load heat runs on battery chargers shall be continued until steady final temperatures have been attained in all parts. During the heat run, thermometers shall be read every half hour, commutation observed frequently to see that it remains

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uniformly good throughout the run, and readings of the shunt field current and voltage required to maintain normal voltage recorded hourly. When the generator reaches a constant temperature elevation, the set shall be shut down and final temperatures taken including commutator surface, armature core, and windings. Allowable temperature increases above ambient temperature for operation at the normal rated output shall not exceed those outlined in NEMA standards. In determining temperature rise, no correction shall be made for barometric pressure, humidity, or for any deviations of the recorded ambient temperature from the standard ambient temperature of reference.

4.6.5 *Dielectric test.* The dielectric test shall be made after all other tests have been completed. The dielectric test shall be made by applying a 60-cycle potential of 1000 volts plus twice the rated voltage of the generator between each individual winding and the frame. Armature and field windings of 35 volts or less shall be tested with 500 volts. Voltages shall be applied for a period of not less than one minute.

5. PREPARATION FOR DELIVERY

For civil agencies, the level of preservation, packaging and packing used are defined in Fed. Std. No. 102.

5.1 Preservation, packaging, and packing.

5.1.1 Motor-generators shall be preserved, packaged, and packed in accordance with MIL-P-16298. Packaging and packing shall be level A, B, or C, as specified in the invitation for bids (see 6.2).

5.2 Marking for shipment.

5.2.1 *Civil agencies.* Marking for civil agencies shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* Shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Motor generators described herein are intended for use in recharging industrial storage batteries which furnish electric power for propulsion and general functioning of materials-handling equipment.

6.2. **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Type and size of charger required. (1.2.1).

(*c*) When a preproduction model is required. (3.2.1).

(*d*) When supply voltage is other than specified. (3.3.1.4).

(*e*) Type of motor required when other than specified. (3.3.2).

(*f*) Special charging connectors, when required. (3.4.7.1).

(*g*) Color of finish coat when other than specified. (3.8.1 and 3.14.1).

(*h*) Additional distribution of technical manuals, when required. (3.13.1).

(*i*) Administrative provisions for inspection records. (4.1).

(*j*) Level of preservation, packaging, packing, and marking required. (5.).

6.3 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail:

Generators or Parts thereof, NOIBN.
Carload minimum weight 30,000 pounds.

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Motor:

Generators, or Parts thereof, NOI.
Truckload minimum weight (w) 30.6 pounds, (w) subject to Rule 115, National Motor Freight Classification.

6.4 Information to accompany bid. Contracting officers should require each bidder to submit a positive statement as to whether or not the item offered fully conforms to the terms of this specification. In addition, each bidder should furnish complete descriptive literature, drawings or photographs, and technical data covering the equipment he proposes to furnish, identifying the equipment by the name of the manufacturer and the model number. Lack of specific and incomplete information will be sufficient cause for rejection of bid(s). Where the bidder's product differs from the specific requirements, each point of difference should be clearly described. This requirement is set forth to facilitate the review of bids and is not to be construed by the bidder as waiving any of the requirements of the specification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—Q SIG T E

Navy—S Or Y

Air Force—SAAMA

Copies of this specification may be purchased for 10 cents each.

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